1/5 Study of Life

This lecture is about three big things:

1. How this course thinks about **science**
2. What we mean by **evolution** and **ecology**
3. What it actually means for something to be **alive**

Everything else we do this quarter builds on these ideas.

Every lecture starts with a warm-up question. These are not graded. I'm not judging you. I'm trying to see how hundreds of brains are thinking at once.

Second slide: **learning objectives**. These matter. They tell you what you're expected to *do* with the material—not just what you've heard. If it's on the learning objectives, it's fair game for exams.

Community Agreement

Here's how we operate together:

- Be respectful. No lying. No cheating.
- Be mentally present. Laptops are fine—doomscrolling is not.
- Be willing to be wrong.

Science is built on being wrong and figuring out why. Some of the most important discoveries we'll talk about came from people being *confidently incorrect*. That's where the interesting stuff happens.

When I ask, **"Does that make sense?"**, I mean it. If it doesn't, say so. Loudly. Science does not advance by silent confusion.

The Process of Science

You've seen this before, but we're doing it carefully.

1. **Observation** – You notice something about the world.
2. **Question** – Why is that happening?
3. **Hypothesis** – A testable explanation.
4. **Prediction** – What should happen *if* that hypothesis is correct?
5. **Experiment / Data** – Test it.
6. **Explanation** – Interpret results.
7. **Repeat** – Because good studies generate more questions than answers.

A key concept here: **the null hypothesis**. The null hypothesis says *nothing is happening*. No effect. No relationship. Flat line.

Biking in Cold Example:

If we ask whether air temperature affects biking speed, the null hypothesis is *air temperature has no effect on biking speed*. Not "you bike the same speed every day," but "temperature isn't the reason speed changes."

Alternate hypotheses must be:

- About the correct variables
- Measurable
- Directional when possible

Words like *good* or *bad* are useless in science. Numbers win.

From Hypotheses to Theories

One experiment does not make a theory. A **theory** is built from:

- Many hypotheses, experiments, scientists, decades, etc…

Which brings us to evolution.

Who Developed the Theory of Evolution?

Short answer: **not one person**. Yes, Charles Darwin matters—but he stands on a mountain of earlier thinkers.

- **Aristotle** argued species were fixed and unchanging.
    - Early philosophers questioned that idea centuries before Darwin.

- **Alexander von Humboldt** noticed patterns in climate and biodiversity across continents.
- **Charles Lyell** showed that slow geological processes, over immense time, create massive change.
- **Mary Anning** uncovered fossils that made extinction undeniable.

Darwin enters this story with two critical ingredients:

1. Massive global observation (Voyage of the Beagle)
2. A framework for how populations change over time

He didn't have genetics. He didn't know about DNA. So he tested ideas using pigeon breeding—and discovered that hidden variation can reappear. Then **Alfred Russel Wallace** independently reached the same conclusions. They published together. Important points:

- At that stage, evolution was a **hypothesis**, not a theory.
- It took until the **1930s Modern Synthesis**—combining genetics, statistics, population biology, and evolution—for what we now call the **theory of evolution** to exist.

The takeaway: No one person "creates" a scientific theory. Science is collaborative, slow, and messy.

What Is Life?

Biology is the study of life, so we need criteria. Living things…

- Are organized (anatomy wise)
- Process energy
- Grow and develop
- Reproduce
- Respond to stimuli
- Regulate internal condition (homeostasis)
- Evolve at the population level

Redwood trees do all of this. Humans do all of this. Bacteria do all of this. Fire does *some* of this—but not all. Fire instead:

- Responds to stimuli
- Uses energy
- Can spread (reproduce)

But it is **not cellular**. The smallest unit of life we recognize is the **cell**. This boundary is imperfect—viruses make everything uncomfortable—but biology lives in these gray zones. Get used to that.

Ecology and Evolution

These are not separate ideas.

- **Ecology** = interactions and relationships
- **Evolution** = genetic change across generations
    - **Microevolution**: small changes, short timescales
    - **Macroevolution**: large changes, long timescales
    - Examples: Mutation
- Terms that fall under both:
    - Natural selection, Mutation, Phenotype, Competition, Mutualism, Mate choice

And here's the key idea that explains why this course exists:

> *Nothing in biology makes sense except in the light of evolution.*
>
> *Nothing in evolution makes sense except in the light of ecology.*
>
> *Nothing in either makes sense without the other.*

Interactions drive genetic change. Genetic change alters interactions. You cannot separate them.

Key Takeaways

1. Biology is not a collection of facts — it's a way of asking and testing questions.

Everything in this course flows from the scientific process: observation → hypothesis → prediction → testing → revision. If you skip the logic and memorize conclusions, you will get lost fast.

2. A null hypothesis always assumes no effect or no relationship.

If you can't clearly state the null, you don't understand the question yet. This shows up constantly in experiments, exams, and data interpretation.

3. Hypotheses must be measurable and specific.

Words like *good*, *bad*, *healthy*, or *better* are red flags unless they're explicitly quantified. Biology cares about direction and magnitude, not vibes.

4. One experiment never proves a theory.

Scientific theories (like evolution) are built from decades of evidence across many fields. If you think "theory = guess," you're already wrong.

5. Evolution was not developed by one person.

Darwin matters, but evolution is the product of many thinkers, many datasets, and many corrections over time. Science is collaborative and cumulative.

6. Evolution describes population-level genetic change, not individual change.

Individuals do not evolve. Populations do. If you say “this organism evolved during its lifetime,” that’s a hard conceptual error.

7. Life is defined by a set of properties — and the boundaries are messy.

Cells are the smallest unit of life *by definition*, but biology is full of edge cases (yes, viruses). Expect gray areas, not clean boxes.

8. Ecology = interactions. Evolution = change. They are inseparable.

Interactions drive selection. Selection alters interactions. If you try to understand one without the other, your explanation will be incomplete.

9. Being wrong is not failure — it’s normal science.

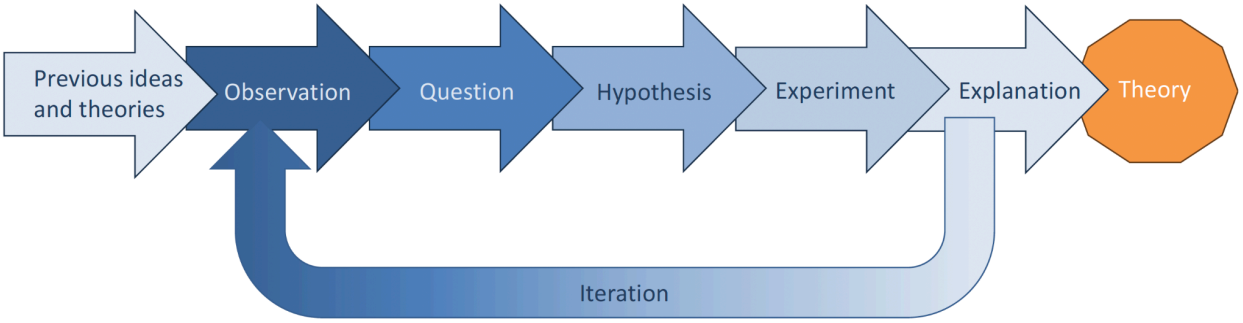
Many major discoveries came from confident mistakes. What matters is whether the mistake leads to better questions.

10. If something doesn’t make sense now, it probably won’t make sense later unless you fix it.

This course builds vertically. Confusion compounds. Speak up early.

Learning Outcomes:

Describe the process of science from observation to theory, including how multiple lies of evidence can be integrated and the importance of iteration and repetition



The process of science is an iterative journey that transforms initial observations into robust theories through rigorous testing and the integration of diverse evidence.

From Observation to Hypothesis

- The scientific process typically begins with an **observation**, which leads to a specific **question** about why a phenomenon is occurring or what factors are influencing it. To answer this question, researchers formulate a **hypothesis**, which is a proposed explanation that predicts a possible outcome. Scientific inquiry generally utilizes two types of hypotheses:
    - **Null Hypothesis:** This predicts that there is **no relationship** between the variables being tested and that the outcomes of the control and treatment groups will be the same.
    - **Alternate Hypothesis:** This predicts that a relationship **does exist** and that the variables will produce different outcomes. For a hypothesis to be effective, it must be **quantifiable and measurable**, avoiding subjective terms like "good" or "bad" in favor of specific directions such as "increasing" or "decreasing".

Experimentation and the Role of Iteration

- Once a hypothesis is established, scientists design an **experiment** to test it, subsequently developing an **explanation** based on the resulting data. A critical aspect of science is **iteration**; a single study rarely provides a final answer and often generates more new questions than it solves. This creates a cycle where explanations lead to further observations and refined hypotheses. This repetitive process is essential for building a **robust understanding** of a system.

Integrating Evidence into Theory

- A **theory** is not merely an untested idea or a single research paper; rather, it is a comprehensive framework developed after the scientific process has been repeated enough times to establish a high level of certainty. The **theory of evolution** serves as a primary example of how multiple lines of evidence are integrated over time. It draws from various disciplines and contributions, including:
- **Biogeography:** Observations of how similar climates on different continents produce different biological outcomes.
- **Geology and Paleontology:** Evidence from the fossil record and the understanding that slow, long-term processes can result in major changes.
- **Genetics:** The integration of **Mendelian inheritance** and modern understanding of DNA to explain how traits are passed down.
- Ultimately, a theory is a communal achievement; even when named after an individual like Charles Darwin, it represents **decades or centuries of work** by a vast community of scientists across multiple fields who test and refine the underlying hypotheses.

Analogy for Understanding:

- Think of the scientific process like **building a high-resolution mosaic**. An initial observation is a single tile. The hypothesis is your plan for where other tiles might go. Iteration is the act of trying different tiles to see if they fit, often realizing you need to change your pattern as the picture grows. The final theory is the complete, detailed image that only becomes clear after thousands of tiles (lines of evidence) have been placed by many different people over time.

List the traits that all living things exhibit and describe examples of how living things meet each requirement (and how non-living things fail to meet one or more requirements)

1. **Order and Organization:** Living things have a high degree of structure. For example, **humans** exhibit organization with heads and feet at opposite ends and organs consistently placed in specific locations. Similarly, **coastal redwoods** display organization through distinct tissues and a structure with roots at the bottom and leaves at the top.
2. **Energy Processing:** Organisms must take in and use energy to fuel their activities. **Humans** process energy by consuming food, such as raw fish. **Trees** meet this requirement through the process of **photosynthesis**.
3. **Growth and Development:** All living things increase in size or change over their lifespan. **Humans** are born significantly smaller than their adult forms, and **trees** grow from tiny seeds into massive structures. This trait can also include complex changes like metamorphosis.
4. **Evolutionary Change:** While individuals do not evolve during their lifetimes, **populations** experience genetic changes over successive generations. Evidence for this is seen in the **fossil record**, such as the changing skull shapes of extinct human relatives or fossilized trees that show how species adapted over time.
5. **Regulation (Homeostasis):** Living things must maintain a stable internal environment. **Humans** regulate body temperature through adaptations like **sweating**. **Plants** regulate their internal water balance by opening or closing pores in their leaves to manage the trade-off between CO2 intake and water loss.
6. **Reproduction:** Organisms produce offspring to continue their species. **Humans** reproduce sexually, while **trees** produce seeds to start the next generation.
7. **Response to Stimuli:** Living things react to their environment. In **humans**, body temperature and blood flow change in response to emotions, such as **blushing** when embarrassed. **Trees** respond to seasonal changes, and some California conifers have adapted to fire by only opening their cones to release seeds after being heated by a blaze.

Non-Living Things and the "Cell" Requirement

Non-living things are excluded from the definition of life because they fail to meet the full suite of these traits and lack a cellular structure. The sources specify that the **cell is the smallest unit of life**; therefore, anything smaller than a cell, such as an atom, is not alive.

Fire is a primary example of a non-living entity that can be deceptive because it appears to meet several requirements:

- **Response to Stimuli:** A flame moves when blown on or changes color if different materials are added.
- **Reproduction:** Fire can "reproduce" by spreading from one object to another.
- **Energy Processing:** Fire consumes fuel and releases energy.

Despite these similarities, **fire is not alive** because it is not composed of cells and does not meet the biological requirements for organization or genetic evolutionary change.

Analogy for Understanding: Think of the properties of life like a **seven-digit security code** required to open a "Life" vault. A non-living thing like fire might correctly enter three or four of the numbers (reproduction, energy, response), but because it lacks the other digits—especially the **cellular "key"**—the vault remains locked, and it cannot be classified as a living organism.

Explain why ecology and evolution are taught together in a single course

1/7 Coral Bleaching (Discussion)

Gradescope:

Which outcome do you predict?

1/7 Biodiversity

Biodiversity refers to the variety of life, but it is not a single, simple concept. It exists at multiple biological scales, ranging from differences among genetically identical individuals to variation across entire ecosystems and taxonomic groups. One of the most common ways biologists organize biodiversity is through **taxonomy**, which groups organisms into nested categories such as species, genus, family, order, class, phylum, and kingdom. Humans, for example, belong to the species *Homo sapiens*, within the genus *Homo*, family Hominidae, order Primates, class Mammalia, phylum Chordata, and kingdom Animalia. These categories are nested, meaning each higher level contains all the diversity of the levels below it. Importantly, higher taxonomic ranks do not need to contain large numbers of species; some phyla, such as Ginkgophyta, contain only a single living species, while others, like Chordata, contain tens of thousands. Taxonomic rank reflects evolutionary relationships, not an expectation of diversity.

When biologists talk about biodiversity, they may be referring to **genetic diversity** within a species, species diversity within a community, diversity across higher taxonomic levels, or functional diversity, which describes the roles organisms play in ecosystems. In this lecture, the primary focus is on species diversity, defined as the variety of species occupying the same place at the same time. Species diversity has two key components: species richness, the total number of species present, and species evenness, the relative abundance of those species. An ecosystem with many species dominated by one extremely common species is less diverse than an ecosystem where species are more evenly represented. Both richness and evenness must be considered to meaningfully compare ecosystems.

A major challenge in understanding biodiversity is that human intuition is often misleading. Groups that people are most familiar with, such as mammals, are not necessarily the most diverse. Mammals include roughly 5,500 species, whereas insects include about one million described species, with beetles alone accounting for hundreds of thousands. Ray-finned fishes, fungi, and even ferns all surpass mammals in species richness. This highlights an important principle in biology: personal experience does not accurately reflect global diversity patterns.

Biodiversity is also unevenly distributed across Earth. Certain regions, known as biodiversity hotspots, contain exceptionally high numbers of endemic species—species found nowhere else—and have experienced extensive habitat loss. California qualifies as a biodiversity hotspot

due to its large number of endemic plant species combined with widespread habitat destruction from agriculture and urban development. Although hotspot designations often focus on plants, plant diversity is strongly linked to overall biodiversity because plants form the energetic foundation of ecosystems through photosynthesis.

Large-scale patterns of biodiversity are shaped by geological history, climate, and physical barriers. Continental drift has played a major role by separating landmasses that were once connected, isolating populations and allowing them to evolve independently. This process explains distributions such as penguins across southern hemisphere continents and the distinct evolutionary history of India, which originated as part of Gondwana before colliding with Asia. Sea level changes further influence biodiversity by periodically connecting and isolating landmasses, particularly in island systems. However, some barriers, such as deep ocean trenches, have never been exposed, preventing migration even during periods of low sea level. Alfred Russel Wallace identified one such barrier, now known as Wallace's Line, which separates distinct biological communities despite close geographic proximity.

Climate is another major driver of biodiversity, especially through its influence on temperature, sunlight, and precipitation. The equator receives the most direct solar energy because the Earth is round, leading to higher temperatures, greater evaporation, and increased rainfall. These conditions support high primary productivity and, consequently, high biodiversity, which is why tropical rainforests are among the most diverse ecosystems on the planet. On a regional scale, physical features such as mountain ranges shape biodiversity through rain shadow effects. As moist air rises over mountains, it cools and releases precipitation on the windward side, leaving the leeward side dry. In California, Pacific air masses lose moisture over both the Coast Ranges and the Sierra Nevada, resulting in wet western regions and extremely dry areas east of the Sierra Nevada.

Overall, biodiversity patterns reflect the combined effects of evolutionary history, climate, geography, and ecological interactions. Regions with abundant water and energy tend to support more diverse plant communities, which in turn support greater diversity in animals, fungi, and other organisms. There are exceptions, such as reptiles, which often thrive in hot, dry environments, and freshwater fishes, whose diversity depends on aquatic rather than terrestrial conditions. Understanding biodiversity therefore requires moving beyond intuition and using ecological and evolutionary principles to explain why life is distributed the way it is across the planet.

Key Takeaways

- **Biodiversity operates at multiple scales**: It includes genetic variation within species, differences among species, higher taxonomic diversity, and functional diversity tied to ecological roles. No single metric captures it all.
- **Taxonomy reflects evolutionary relationships, not "amount" of diversity**: Higher taxonomic ranks don't need many species; what matters is shared ancestry, not species count.
- **Species diversity has two components**:

    - **Species richness** counts how many species exist; **species evenness** describes how evenly individuals are distributed among those species. You need both to compare ecosystems honestly.
- **Human intuition about diversity is unreliable**: Familiar groups like mammals are relatively species-poor compared to insects, fungi, fishes, and even plants.
- **Biodiversity is unevenly distributed geographically**: Biodiversity hotspots contain many endemic species but are also heavily threatened by habitat loss.
- **Plants underpin biodiversity patterns**: Because they capture energy via photosynthesis, plant diversity strongly predicts diversity at higher trophic levels.
- **Geological history shapes global biodiversity**: Continental drift, sea-level change, and long-standing physical barriers isolate populations and drive independent evolution.
- **Physical barriers matter—even small ones**: Some barriers (e.g., Wallace's Line) prevent species mixing despite close proximity, leading to sharp biological divides.
- **Climate strongly influences biodiversity**: High energy and water availability—especially in the tropics—support high productivity and high species diversity.
- **Regional climate patterns matter too**: Features like mountain ranges create rain shadows that dramatically restructure ecosystems over short distances.
- **Big picture**: Biodiversity patterns emerge from the interaction of evolution, climate, geography, and ecology—not from what humans happen to notice most easily.

Learning Outcomes:

Describe how biodiversity can be measured at different scales

Describe patterns of species diversity across the tree of life

Describe patterns of species diversity across geographic space

List and describe factors that influence biodiversity across

1/9 Climate & Climate Change

Today we're shifting gears slightly to focus on **climate and climate change**, building directly on what we discussed last time about biodiversity. Specifically, we're going to zoom in on the **abiotic side of the system**—the physical structure of the Earth and how it shapes environmental conditions and, ultimately, biomes.

A major tool we'll be using today is something called a **climatograph**, and you're going to see a lot of these—because, frankly, I love them, and they're incredibly useful. A climatograph shows you months of the year on the x-axis. On the left y-axis, you'll see the average **monthly temperature** in degrees Celsius, shown as a gray line. On the right y-axis, you'll see **precipitation**, measured in millimeters, shown as black bars. These data come from a global climate database called *WorldClim*, which aggregates long-term climate records for locations around the planet.

When you look at a climatograph, you should immediately be asking a few questions. First: is this place generally **wet or dry**? Second: is temperature relatively constant, or does it show strong **seasonality**? Third: if both temperature and precipitation are seasonal, **do the warm and wet seasons line up—or not**? Those three questions alone will get you surprisingly far in identifying biomes.

Let's start by clarifying a really important distinction: **weather versus climate**. Weather is what's happening right now—today's temperature, whether it's raining, whether it's cloudy. Climate, on the other hand, is about **long-term averages and patterns**: mean annual temperature, average precipitation, typical seasonality. So if I show you a map of California's *average annual rainfall*, that's climate. If I show you a map of this morning's temperatures, that's weather. The keyword here is *average*.

Now, when we talk about biomes, you'll quickly discover there's no single "correct" list. That's because biomes don't exist as sharp categories in nature—they exist along **gradients**. Deserts gradually transition into shrublands, which transition into grasslands, and so on. For today, we're going to use **broad biome categories**. That means we're lumping together things like arid deserts, xeric shrublands, and dry steppes under the general label "desert," because climatographically, they look very similar.

One useful way to visualize biome patterns globally is to plot **temperature against precipitation**. If a place is extremely cold and extremely dry, you expect tundra. If it's warm and extremely wet, you expect tropical rainforest. But temperature and precipitation aren't the whole story. **Disturbance**, especially fire, also matters. Some places could support forests based on climate alone, but frequent fire keeps them as shrublands or grasslands. Keep that in the back of your mind—we'll come back to it later in the course.

Let's look at a classic example: a **tropical rainforest climatograph**. You'll see tall precipitation bars year-round and a nearly flat temperature line. That flat line matters. Tropical rainforests are essentially the *only* biome we cover that shows almost no temperature seasonality, because they sit near the equator and receive consistent solar energy year-round.

Now contrast that with a **temperate grassland**. Here, precipitation is lower overall and clearly seasonal. Temperature also varies dramatically—often dipping below freezing in winter and climbing above 25°C in summer. Importantly, in grasslands, the **warm season is also the wet season**. That timing is one of your biggest clues.

A lot of biome identification comes down to ruling things out. If there's significant rainfall, it's not a desert. If temperatures drop below freezing, it's not a tropical rainforest. If precipitation is strongly seasonal, it's not a temperate deciduous forest, which tends to have rainfall spread fairly evenly across the year. Forests need more water than grasses—so if precipitation is lower and seasonal, grassland becomes much more likely.

Mediterranean ecosystems—like where we live—are a special case. They have **seasonality in both temperature and precipitation**, but critically, the **warm season is dry and the cool season is wet**. That's the defining feature. If the warm season is wet, you're looking at grassland. If the warm season is dry, you're looking at Mediterranean climate. That single timing difference matters a lot.

Now let's talk hemispheres. You may notice that two deserts—the Sahara and the Atacama—have very different temperature patterns, even though both are extremely dry. That difference comes from **which hemisphere they're in**. The Earth's axis is tilted, not straight up and down. That tilt—not distance from the Sun—is what causes seasons. When the Northern Hemisphere is tilted toward the Sun, it's summer here and winter in the Southern Hemisphere. Six months later, that flips.

This matters because climatographs can tell you which hemisphere a biome is in, based on *when* temperatures peak. But—and this is important—always read the axes. If I really wanted to be mean, I could label the x-axis July to June instead of January to December. I probably won't do that on an exam. Probably.

So far, we've focused on climate patterns as drivers of biomes. Now we're going to pivot to **climate change**, specifically the evidence that it is **anthropogenic**, meaning human-driven.

The Earth has a **radiation budget**. Solar energy comes in, some is reflected back to space, some is absorbed by the atmosphere, and some is absorbed by the Earth's surface. Greenhouse gases—like carbon dioxide, methane, and nitrous oxide—trap a portion of that energy, keeping the planet warm enough to be livable. That process itself is not bad. Without it, Earth would be frozen.

The problem is **rate and magnitude of change**. Concentrations of these gases are increasing rapidly. Carbon dioxide has risen from about 320 parts per million in the mid-20th century to over 420 parts per million today. Ice core data allow us to look back 800,000 years, and never during that time did CO_2 exceed 300 ppm—until now. This increase is not subtle. It's dramatic.

The most direct consequence is rising temperature. Since 1979, **every single year** has been warmer than the historical average. If you were born after 1979, you have never experienced a below-average year. If you were born after February 1985, you have never experienced a below-average month. The ten hottest years on record are the ten most recent years. That's not coincidence—that's a trend.

Warming is not uniform across the globe. The Arctic is warming much faster than lower latitudes. And warming doesn't just affect temperature—it affects **precipitation**. Precipitation patterns are more complex, but the general rule seems to be: **dry regions get drier, wet regions get wetter**.

In California, this plays out seasonally. Wet seasons are projected to get wetter; dry seasons are projected to get drier. In a Mediterranean climate—where the dry season is already the hot season—that combination is particularly dangerous. The obvious consequence is **fire**.

Since records began in 1932, **18 of the 20 largest fires in California history have occurred since 2000**. Many of them since 2020 alone. This isn't just variability. It's consistent with rising temperatures and changing precipitation patterns.

So when we talk about climate change, we're not just talking about abstract global averages. We're talking about real ecological consequences—shifts in biome boundaries, increased disturbance, and conditions that challenge the organisms already living there.

Alright. With that, I'll see you on Monday. Enjoy your weekend.

Key Takeaways

- **Climate vs. weather is a scale difference**: Weather describes short-term conditions; climate describes long-term averages and patterns. If it's averaged, it's climate.
- **Climatographs are biome-identification tools**: Read precipitation amount, temperature seasonality, and whether warm and wet seasons align to infer biome type.
- **Biomes exist along gradients, not hard boundaries**: Broad categories are simplifications; real ecosystems transition gradually and overlap.
- **Temperature and precipitation jointly shape biomes**: Extremely cold/dry environments produce tundra; warm/wet conditions produce tropical rainforests—but disturbance also matters.
- **Tropical rainforests are climatically distinctive**: They show high year-round precipitation and almost no temperature seasonality due to constant solar input.
- **Timing of wet vs. warm seasons is critical**: Grasslands have warm–wet seasons; Mediterranean climates have warm–dry and cool–wet seasons.
- **Forests require more consistent water than grasslands**: Year-round precipitation distinguishes temperate forests from seasonally dry grasslands.
- **Hemispheres flip seasonal patterns**: Earth's axial tilt—not distance from the Sun—causes seasons, and climatographs can reveal hemisphere location if axes are read carefully.
- **Greenhouse gases are necessary but increasing too fast**: CO_2, methane, and nitrous oxide trap heat naturally, but their rapid rise is disrupting Earth's energy balance.
- **Modern CO_2 levels are historically unprecedented**: Ice core records show current concentrations far exceed anything seen in the past 800,000 years.
- **Global temperatures show a clear upward trend**: Every year since 1979 has been warmer than the long-term average; recent decades dominate record highs.

- **Climate change impacts are uneven**: The Arctic is warming fastest, and precipitation patterns are shifting unevenly across regions.
- **California is especially vulnerable**: Hotter dry seasons and wetter wet seasons intensify fire risk in an already fire-prone Mediterranean ecosystem.
- **Recent wildfire patterns reflect climate change, not chance**: The majority of California's largest fires have occurred since 2000, aligning with warming and drying trends.
- **Climate change reshapes ecosystems**: Beyond averages, it alters biome boundaries, disturbance regimes, and the survival conditions for organisms.

Learning Outcomes:

Distinguish weather and climate and describe how both are driven by the Earth's energy budget

Interpret climatographs to identify and compare precipitation, temperature, and seasonality between geographic locations

Describe the evidence that Earth's climate is experiencing anthropogenic-driven change